

Predictive Demand Modelling for Mobility-as-a-Service (MaaS) Using Machine Learning: A Smart City Framework and Its Transferability to Rapidly Urbanising African Cities

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1. Abstract

Urban mobility systems in rapidly growing cities face a fundamental tension between rising passenger demand and the structural limitations of existing infrastructure. Mobility-as-a-Service (MaaS), a transport model that integrates multiple modes under a single digital platform, offers a potential solution. However, the operational viability of any MaaS platform depends critically on the accuracy of its demand forecasting capability. Without reliable predictions of how many users will require which modes, where, and when, operators cannot allocate vehicles effectively, and the promise of seamless integrated mobility cannot be delivered.

This research proposes to develop and validate a machine learning framework for multimodal MaaS demand prediction, using open-access trip data from New York City as the primary empirical testbed. The framework evaluates four model architectures: autoregressive integrated moving average (ARIMA) baselines, gradient boosting ensembles (Chen & Guestrin, 2016), long short-term memory networks (Hochreiter & Schmidhuber, 1997), and spatial-temporal graph neural networks (Li et al., 2018). Model outputs will be interpreted using SHAP values (Lundberg & Lee, 2017) to identify the most influential predictors of demand volatility across transport modes.

A critical secondary contribution is a structured transferability analysis examining the conditions under which the proposed framework can be adapted to data-scarce environments in Sub-Saharan Africa, using Lagos, Nigeria as the reference context. Lagos, home to an estimated 20 million residents, relies on an informal, undigitised transport ecosystem that has no equivalent predictive infrastructure (Ibitayo, 2012; Buhari et al., 2020). This research argues that methodological advances developed in data-rich cities can and should be adapted to serve the planning needs of cities like Lagos, where the consequences of poor transport management fall heaviest on the urban poor (Castro et al., 2022). The study carries implications for MaaS policy, smart city investment, and AI-driven transport governance in the Global South.

2. Background and Motivation

The world is in the middle of an urban mobility crisis. By 2050, approximately 68% of the global population will live in cities, with the fastest growth concentrated in Sub-Saharan Africa and South Asia (United Nations, 2018). In Nigeria, urban population growth has accelerated from 10.2% of the total population in the early 1950s to projections exceeding 68% by the middle of this century, with Lagos growing at between 5 and 8 percent annually to become one of the 15 largest urban agglomerations in the world (Badejo, 2013). The transport infrastructure of these cities has not kept pace. In Lagos, an estimated 60% road network congestion and an average vehicle speed of barely 17.2 kilometres per hour have earned it recognition as one of the worst cities in the world for drivers (Adagunodo & Filani, 2023; Castro et al., 2022).

Against this backdrop, MaaS has emerged as one of the most actively discussed frameworks for rethinking urban mobility. Hensher (2017) described MaaS as a digital-age reimagining of the public transport contract, one where the passenger receives individualised,

multimodal travel options through a single platform rather than navigating fragmented, mode-specific systems. Jittrapirom et al. (2017) offered a comprehensive definitional review, identifying key MaaS integration tiers ranging from simple multimodal information to fully bundled subscription services. The concept has moved from theory to trial in Helsinki, Antwerp, and Sydney (Wong et al., 2020).

The technical backbone of any functioning MaaS platform is its demand intelligence layer. Knowing in advance where passengers will originate, what modes they will prefer, and when peak demand will occur allows operators to pre-position vehicles, dynamically price journeys, and maintain service quality. Classical transport demand models, which rely on aggregate zone-level travel surveys and fixed trip-generation assumptions, are ill-suited to this real-time, disaggregated need. The maturation of machine learning methods for spatiotemporal prediction offers an alternative approach that is both more accurate and more responsive to real-world conditions.

The motivation for connecting the technical work to Nigeria is substantive, not symbolic. Lagos is making credible investments in formal public transport, including Bus Rapid Transit, light rail, and inland waterway services (Buhari et al., 2020). As these systems expand and generate digital trip records, the absence of demand forecasting infrastructure becomes an increasingly costly planning gap. Transport decisions taken without predictive intelligence tend to reproduce the reactive, supply-driven approach that has historically failed Nigerian cities. This research is designed, from the outset, to produce outputs that are relevant to that planning challenge.

3. Problem Statement

The literature on transport demand forecasting contains a well-documented and persistent gap: there is no validated, multimodal, spatially granular prediction framework designed for the operational requirements of integrated MaaS platforms. Three specific limitations characterise the current state of the field.

First, existing predictive models are predominantly unimodal. Most machine learning studies in urban transport target a single mode, typically taxi or ride-hailing demand, without accounting for cross-modal substitution effects. When a metro line faces disruption, demand shifts to buses, taxis, and cycling in patterns that a single-mode model cannot anticipate. The MaaS demand forecasting problem is inherently multimodal, and robust, operationally validated frameworks that explicitly model these interactions remain limited.

Second, current spatial modelling efforts are not designed to capture the complex spatial dynamics of an integrated multimodal network. While recent deep learning architectures (such as spatial-temporal graph neural networks) excel at predicting highly granular, zone-level demand for isolated modes, these models do not translate well to MaaS environments. Aggregate city-level predictions are operationally inadequate if a model cannot identify the uneven spatial distribution of demand across multiple overlapping transit networks. Consequently, the field lacks a spatial framework capable of identifying how localized spatial interactions in one mode affect the repositioning of supply in another.

Third, few studies have systematically examined the conditions under which ML-based MaaS demand frameworks, developed in data-rich cities, can be transferred to data-scarce urban environments in the Global South. The cities that most need advanced transport intelligence are precisely those for which validated frameworks remain scarce, and for which building equivalent

datasets from scratch would be prohibitive. This transferability gap, anchored in the context of Lagos, is the principal research problem this study addresses.

4. Research Questions

This study is guided by three research questions:

1. **RQ1:** How accurately can a machine learning framework forecast multimodal MaaS demand at fine spatial and temporal resolution compared to classical statistical baselines, and which model architecture achieves the best balance between predictive performance and computational cost?
2. **RQ2:** Which exogenous variables, including weather conditions, temporal patterns, public events, and service disruptions, have the greatest influence on MaaS demand variation across transport modes, and how can SHAP-based interpretability methods make these relationships actionable for transport operators?
3. **RQ3:** Under what conditions, and through what methodological adaptations, can a demand forecasting framework validated on open-access data from a data-rich city such as New York City be transferred to an informal, data-scarce urban environment such as Lagos, Nigeria?

These questions form a coherent progression. RQ1 establishes the technical benchmark. RQ2 generates interpretable insights from that benchmark. RQ3 applies those insights to the practical challenge of transferability, which is the bridge between method development and real-world impact in the Nigerian context.

5. Literature Review

5.1 Mobility-as-a-Service: Concept and Operational Requirements

MaaS entered academic and policy discourse primarily through Finnish transport researchers in the early 2010s, with definitional consolidation offered by Jittrapirom et al. (2017) in their critical review of definitions, scheme assessments, and key challenges. They defined MaaS as a mobility distribution model in which a customer's major transport needs are met over one digital interface, encompassing journey planning, booking, payment, and real-time information across all integrated modes. Their typology of MaaS integration levels remains the most cited framework in the field.

Hensher (2017) positioned MaaS as a structural disruption to the public transport contracting model, arguing that a shift from fixed-route operations toward demand-responsive, platform-mediated services would require fundamentally different operational logic, specifically the capacity to act on accurate real-time demand signals. Wong et al. (2020) extended this by proposing a modal efficiency framework linking land use to MaaS service design, arguing that which modes to integrate and at what intensity cannot be answered without reliable spatial demand intelligence. Together, these works ground the technical problem of this research directly in MaaS governance theory.

5.2 Classical Transport Demand Forecasting Methods

The traditional four-step transport demand model sequentially estimates trip generation, distribution, modal split, and route assignment using aggregate zone-level data. While foundational for strategic planning, it operates on horizons of years or decades and is not designed for real-time operational decisions. ARIMA and its seasonal variant SARIMA have been applied to short-term ridership forecasting, but both assume linearity and stationarity in the demand signal,

assumptions that break down under the dynamic conditions of modern urban transport systems. These classical approaches serve as the baseline benchmarks in this study, providing the reference point against which machine learning performance improvements are quantified.

5.3 Machine Learning Architectures for Urban Transport Forecasting

Gradient boosting methods, particularly XGBoost (Chen & Guestrin, 2016), build sequential ensembles of decision trees where each tree corrects the residuals of its predecessors. This approach handles nonlinear feature interactions without manual specification and has demonstrated strong performance on tabular transport datasets. Long short-term memory networks (Hochreiter & Schmidhuber, 1997) introduced a gating mechanism into recurrent neural network architecture, allowing the model to learn which past observations are relevant to the current prediction and resolving the vanishing gradient problem that limited earlier recurrent models. LSTMs have since been applied to taxi demand forecasting, metro ridership prediction, and ride-hailing surge modelling with consistent improvements over classical baselines.

The application of graph neural networks to transport forecasting represented a conceptual advance by allowing models to reason about network topology rather than treating locations as independent. Li et al. (2018) proposed the Diffusion Convolutional Recurrent Neural Network (DCRNN), modelling traffic flow as a diffusion process on a directed graph using an encoder-decoder architecture with scheduled sampling to capture both spatial dependency and temporal sequence simultaneously. Their model achieved 12 to 15 percent improvement over state-of-the-art baselines on two large-scale road network datasets. Despite these advances, the literature remains predominantly mode-specific and has not produced a validated multimodal MaaS demand forecasting framework.

5.4 SHAP Values and Interpretable Machine Learning in Transport

A persistent critique of machine learning in transport planning is the opacity of predictions. Lundberg and Lee (2017) addressed this by introducing SHapley Additive exPlanations (SHAP), a framework grounded in cooperative game theory that assigns each input feature a marginal contribution to each individual prediction. SHAP values satisfy the properties of local accuracy, consistency, and missingness, which earlier feature importance methods did not. The framework has been widely adopted as the standard for interpretable machine learning in applied sciences. In this research, SHAP is the mechanism by which technical model outputs are translated into operational intelligence for transport planners, answering questions such as which weather thresholds reliably suppress cycling demand or which temporal patterns most strongly predict bus demand surges on weekday mornings.

5.5 Urban Transport Challenges in Lagos and the Case for Transferability Research

Lagos presents one of the most acute urban transport challenges in the world. Ibitayo (2012) documented how commercial vehicles operating as the dominant informal transit mode, combined with poor enforcement of traffic regulations, produced feedback loops of congestion and breakdown that degraded service quality and commuter experience. Buhari et al. (2020) found that Lagos state transport interventions have structurally failed to keep pace with population growth, leaving the urban mobility system unable to meet the demand placed on it. Castro et al. (2022) quantified the welfare effects of this failure, showing that transport poverty in low-income Lagos communities produces measurable social exclusion through lost access to employment, education, and essential services.

Conducting predictive transport research in cities like Lagos is constrained fundamentally by data scarcity. The informal nature of danfo minibuses and keke tricycle operations means that origin-destination data, trip timing, and ridership counts do not exist at the scale required to train

machine learning models. This constraint is not unique to Lagos. Researchers addressing this challenge have increasingly adopted framework transfer methodology, in which models trained on open-access data from cities with mature data infrastructure are systematically evaluated for applicability to data-scarce environments. This is the conceptual basis for the Nigerian contribution of this research.

6. Research Methodology

This study adopts a quantitative, computational research design based on secondary data analysis. The methodology proceeds through five sequential phases.

6.1 Data Acquisition and Preparation

The primary dataset will be drawn from the New York City open data ecosystem, chosen for its scale, documentation, and established use in peer-reviewed transport AI research (Li et al., 2018). NYC represents one of the most data-rich urban mobility environments globally. Five datasets will be used:

- NYC Taxi and Limousine Commission Trip Record Data (2021-2023): pickup and dropoff coordinates, timestamps, trip distances, and fares, providing the primary demand signal for road-based for-hire vehicle modes (NYC TLC, 2023).
- Citi Bike NYC System Data: station-level origin-destination records for all bicycle trips, supplying the cycling demand layer of the multimodal framework (NYC Citi Bike, 2023).
- MTA Turnstile Data: hourly entry and exit counts at every New York subway station, constituting the mass transit demand layer (MTA, 2023).

- NOAA Integrated Surface Database: hourly weather observations including temperature, precipitation, wind speed, and visibility, forming the primary exogenous variable set (NOAA, 2023).
- NYC Open Data public events calendar and school term schedules for construction of temporal exogenous features.

Data preprocessing will include outlier detection using interquartile range thresholding, spatial aggregation to the Traffic Analysis Zone (TAZ) level for cross-dataset consistency, temporal resampling to 15-minute intervals, and feature engineering producing lag variables at 1, 2, 4, 8, and 24-hour intervals, rolling statistical summaries, and cyclical time encodings using sine and cosine transformations to preserve circular continuity. The dataset will be partitioned temporally into training (70%), validation (15%), and test (15%) subsets, with the test set comprising the final three months of the study period.

6.2 Exploratory Data Analysis

A thorough exploratory data analysis will characterise demand distributions per mode and per TAZ, apply time-series decomposition to identify trend, seasonal, and residual components, produce heatmap visualisations of spatial demand by time and day, and conduct Pearson correlation analysis between demand and exogenous covariates. Moran's I spatial autocorrelation analysis will quantify spatial dependency in the demand signal and, if significant positive autocorrelation is detected, formally justify the use of graph-based spatial modelling.

6.3 Model Development

Four model classes will be implemented in Python using the scikit-learn, XGBoost, TensorFlow/Keras, and PyTorch Geometric libraries.

ARIMA and SARIMA statistical baselines will be estimated separately per mode using the `pmdarima` library for automated order selection. These univariate models include no spatial or exogenous information and establish the performance floor.

Gradient boosting models, specifically XGBoost (Chen & Guestrin, 2016) and LightGBM, will be trained on a tabular feature matrix including temporal features, spatial zone descriptors, lagged demand values, and all exogenous covariates. Hyperparameter tuning will use Bayesian optimisation via the Optuna framework with 5-fold cross-validation over the training set.

LSTM networks (Hochreiter & Schmidhuber, 1997) will take as input multivariate sequences of 96 time steps, representing a 24-hour look-back window at 15-minute resolution, and produce demand predictions at horizons of 15 minutes, 30 minutes, 1 hour, and 2 hours. Separate LSTM models will be trained per transport mode to allow mode-specific temporal pattern analysis.

A spatial-temporal graph neural network (ST-GNN), drawing on the DCRNN architecture introduced by Li et al. (2018), will use the NYC TAZ adjacency matrix constructed from road network connectivity as the graph structure. Multimodal demand signals will be treated as node features, enabling the model to learn cross-modal spatial dependencies that single-mode models cannot capture.

6.4 Model Evaluation

All models will be evaluated on the held-out test set using three complementary metrics: Mean Absolute Error (MAE), reporting average absolute deviation in interpretable trip units; Root Mean Squared Error (RMSE), applying heavier penalty to large errors relevant to operational peak-demand underprediction; and Mean Absolute Percentage Error (MAPE), providing scale-independent comparison across TAZs with different demand magnitudes. Statistical significance

of performance differences will be assessed using Diebold-Mariano tests at the 5% significance level. Spatial analysis of prediction residuals through choropleth mapping of zone-level MAE will identify geographic areas where performance degrades and guide model refinement.

6.5 SHAP-Based Interpretability Analysis

Following identification of the best-performing model, SHAP values will be computed for the full test set using the framework introduced by Lundberg and Lee (2017). SHAP decomposes each prediction into additive contributions from individual input features using cooperative game theory, producing results that satisfy local accuracy and consistency axioms. The analysis will generate global feature importance rankings per transport mode, local prediction explanations for high-demand events, and interaction plots. Outputs will be synthesised into policy-facing interpretations, identifying which weather thresholds suppress cycling demand, which temporal features predict bus demand surges, and how these relationships vary spatially across zones.

6.6 Transferability Analysis for Lagos, Nigeria

The final methodological phase examines the conditions under which the validated framework can be transferred to Lagos. This proceeds in three stages. First, a comparative characterisation of Lagos and NYC transport data environments will document key structural differences including modal composition, data availability, network topology, and informal sector prevalence. Second, a sensitivity analysis will degrade the NYC dataset to simulate Lagos-like data conditions, reducing spatial resolution, removing mode-specific data layers, and introducing missing data at rates corresponding to estimated Lagos data coverage. Third, transfer learning experiments will assess how much predictive accuracy the ST-GNN retains when pre-trained on NYC data and fine-tuned on limited synthetic Lagos data constructed from Google Maps mobility reports, OpenStreetMap network data, and published academic survey datasets. The outcomes will

define a practical roadmap for deploying demand forecasting in cities where equivalent open data infrastructure does not yet exist.

7. Significance and Impact

This research makes four distinct contributions to knowledge and practice.

The first is theoretical. By framing MaaS demand forecasting as a multimodal, spatially disaggregated prediction task and formally comparing four model architectures against a consistent evaluation protocol, the study fills a gap that the existing literature acknowledges but has not closed. The result is a validated, reproducible methodology rather than a proof-of-concept, which is a necessary precondition for adoption by practitioners.

The second is empirical. The comparative evaluation across ARIMA, gradient boosting, LSTM, and ST-GNN architectures on a multimodal dataset directly addresses the practical question confronting any MaaS operator or smart city planning team: no existing study provides this cross-architecture comparison in a genuinely multimodal framework.

The third is methodological. The SHAP interpretability analysis applied across modes is novel. Existing transport applications of SHAP are predominantly single-mode. Understanding, for example, that rainfall suppresses cycling demand far more than metro demand in the same zone, and that this effect varies predictably with time of day, carries direct implications for how MaaS operators should trigger dynamic supply adjustments.

The fourth and most contextually distinctive contribution is the transferability analysis focused on Lagos. Urban transport research in Sub-Saharan Africa suffers from a persistent

methodological fragmentation: the cities that most need intelligent planning tools are precisely those that data constraints have excluded from the dominant empirical paradigms. This study treats data scarcity not as a reason to avoid the research question but as the research question itself. The findings will be relevant to the Lagos State Government, the Federal Ministry of Works and Housing, the African Development Bank, and development institutions investing in smart city infrastructure across the continent. The study is aligned with UN SDG 11 (Sustainable Cities and Communities), the African Union's Agenda 2063 infrastructure targets, and the International Transport Forum's research programme on data-driven urban mobility.

8. Ethical Considerations

This research will use exclusively publicly available, anonymised datasets. No individually identifiable information will be collected, stored, or processed at any stage. The NYC TLC, Citi Bike, and MTA datasets are published under open data licences and have been used in numerous peer-reviewed studies without ethical restriction. All data will be processed in aggregated form at the Traffic Analysis Zone level. The transferability analysis using synthetic Lagos data will draw only on published academic datasets and publicly available mobility reports. Findings, including model failure modes and the limitations of the transferability analysis, will be reported transparently to ensure that policymakers are not misled about the reliability of AI-driven predictions in contexts where validation data is limited. All code and processed data will be made publicly available on GitHub under an open-source licence.

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